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Received, January 28, 2006.

Accepted, June 27, 2006.

COMPLEMENT ACTIVATION ASSOCIATES WITH SACCULAR CEREBRAL ARTERY ANEURYSM WALL DEGENERATION AND RUPTURE

OBJECTIVE: Saccular cerebral artery aneurysm (SCAA) wall degeneration and inflammatory cell infiltrations associate with aneurysm rupture and subarachnoid hemorrhage, resulting in a devastating form of stroke. The complement system is the key mediator of inflammation and household processing of injured tissue. We studied how complement activation associates with SCAA wall degeneration and rupture to better understand the pathobiology of SCAA wall rupture.

METHODS: Unruptured ($n = 26$) and ruptured ($n = 32$) SCAA fundi resected after microsurgical clipping were studied by immunostaining for complement activation (membrane attack complex [MAC]) and by terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate nick end-labeling reaction for related cell death. Complement activation was correlated with clinical and other histological parameters. Electromicroscopy and immunoelectron microscopy were used for locating MAC depositions at the ultrastructural level.

RESULTS: MAC localized consistently in a decellularized layer in the outer SCAA wall, and was found in all SCAA samples. The percentage of MAC-positive area relative to the total SCAA wall surface area (range, 5–77%) was greater in ruptured ($n = 25$; median, 39%) than in unruptured SCAAs ($n = 18$; median, 20%; $P = 0.005$). It also associated significantly with SCAA wall degeneration ($P < 0.001$), de-endothelialization ($P < 0.001$), and CD163⁺ macrophage ($P = 0.023$) and T-lymphocyte ($P = 0.030$) infiltrations. Apoptotic terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate nick end-labeling-positive nuclei and MAC were located at the same wall areas in four out of 14 double-stained samples, but no double-positive cells were found. Electromicroscopy and immunoelectron microscopy of an unruptured SCAA showed cell death in the MAC-positive layers in the outer SCAA wall.

CONCLUSION: These data suggests that complement activation and MAC formation are involved in SCAA wall degeneration and rupture.

KEY WORDS: Aneurysm, Complement, Inflammation, Intracranial aneurysm, Rupture

Neurosurgery 59:1069-1077, 2006

DOI: 10.1227/01.NEU.0000245598.84698.26

www.neurosurgery-online.com

Saccular cerebral artery aneurysm (SCAA) rupture causes subarachnoid hemorrhage (SAH) with up to 50% mortality (21). Approximately 2% of the population harbor unruptured aneurysms (17), and the general incidence of SAH is approximately 10 per 100,000 people (9). The pathobiology of the SCAA wall needs to be elucidated to better understand aneurysm growth and rupture.

The histological differences in unruptured and ruptured SCAA walls (6, 11) suggest that the SCAA wall is a dynamic structure that

undergoes constant remodeling, which may, in some cases, lead to a more rupture-prone wall structure. Increased inflammatory cell infiltration of the SCAA wall also associates with rupture (6, 11). Activation products of the complement (C) system, the key mediator of inflammation and household processing of injured tissue (15, 20), have been found in the SCAA wall (4).

We hypothesized that C activation is related to SCAA wall remodeling toward a more degenerated and rupture-prone type. Hence,

we studied the deposition of membrane attack complex (MAC), the final product of C activation, in a series of 26 unruptured and 32 ruptured SCAAs and its association with SCAA rupture, wall degeneration, inflammatory cell infiltration, and the patients' clinical findings.

PATIENTS AND METHODS

Patients

A series of 58 patients with 26 unruptured and 32 ruptured SCAAs was studied. All of the SCAA samples originated from different patients. Clinical data were collected from the patients' medical records. SCAA fundus width, length, and neck diameter were measured, as well as the severity of the SAH graded (0–4) according to Fisher et al. (5) from preoperative computed tomographic scans (Table 1).

Microsurgically resected SCAA fundi (JH, MN, AK), distal to the aneurysm clip were snap-frozen in liquid nitrogen or fixed immediately for electron microscopy (EM). Of the studied SCAAs, 43 were previously histologically characterized (6) in continuum with this study. The study was approved by the local ethics committee.

Immunostaining and Terminal Deoxynucleotidyl Transferase-mediated Deoxyuridine Triphosphate Nick End-labeling

The MAC was immunostained with a monoclonal mouse antibody to human SC5b-9 (1:150 dilution; A239; Quidel Corp., San Diego, CA). Immunoperoxidase staining was performed as described (6) with 1% bovine serum albumin in phosphate buffered saline as block and diluent. In the immunofluorescence technique used for double staining and to control the MAC staining pattern, a fluorophore-labeled goat anti-mouse

IgG secondary antibody (1:200 dilution; Alexa fluor 488 or 546; Molecular Probes Inc., Eugene, OR) and 6-diamidino-2-phenylindole counterstain (Vector Laboratories, Burlingame, CA) was used. For double staining, an additional transferase-mediated deoxy-uridine triphosphate nick end-labeling (TUNEL) reaction, after MAC immunofluorescence staining (12), was performed using a fluorescein in situ cell death detection kit (Roche Diagnostics GmbH, Penzberg, Germany).

For negative controls, the primary antibody was omitted or substituted by an irrelevant Ig isotype-matched antibody. Human tonsil and ischemic heart served as positive controls. Snap-frozen intraoperative samples of human brain perforant arteries from temporal lobe resections performed in epilepsy surgery (Department of Neurosurgery, Helsinki University Central Hospital, Helsinki, Finland) were analyzed as normal control arteries. In addition, staining with a mouse monoclonal antibody to human CD59 (1:200 dilution; BRIC 229; IBGRL, Bristol, England) was compared with the MAC staining pattern.

TABLE 1. Distribution of clinical and histological variables from patients with unruptured or ruptured saccular cerebral artery aneurysms

Variables	Bleeding status		P value
	Unruptured aneurysms (n = 26)	Ruptured aneurysms (n = 32)	
Patients			
Age (yr) ^a	55 (28–69)	54 (13–84)	0.563
Females ^b	72% (18 of 25)	52% (16 of 31)	0.120
Patients with multiple (≥2) SCAAs ^b	48% (12 of 25)	26% (8 of 31)	0.085
Patients with previous aneurysmal SAHs ^b	16% (4 of 25)	26% (8 of 31)	0.374
Aneurysms resected for study ^a			
Neck diameter (mm)	4 (2–10)	4 (2–10)	0.951
Fundus width (mm)	6 (3–34)	7 (3–15)	0.115
Fundus length (mm)	7 (3–29)	8 (4–7)	0.114
No. of ruptures	0	1 (1–4)	
Time from rupture to sample resection		14 h (4 h, 47 d)	
Histology of SCAA wall ^{b,c}			
Endothelium (CD31+ cells) totally absent	28% (5 of 18)	61% (14 of 23)	0.035*
Elastic lamina remnants	22% (4 of 18)	13% (3 of 23)	0.438
Intraluminal fresh thrombosis	17% (3 of 18)	44% (11 of 25)	0.059
Intraluminal organizing thrombosis	17% (3 of 18)	56% (14 of 25)	0.009*
Atherosclerotic calcifications	22% (4 of 18)	8% (2 of 25)	0.184
Pads of intimal hyperplasia	44% (8 of 18)	54% (13 of 24)	0.533
Cell proliferation (Ki67+ cells)	17% (3 of 18)	47% (9 of 19)	0.046*
Inflammatory cell infiltrations ^{a,d}			
CD11b+ macrophages	0 (0–37)	2.5 (0–60)	0.018*
CD68+ macrophages	0 (0–16)	0 (0–26)	0.046*
CD163+ macrophages	0 (0–30)	8 (0–38)	0.002*
CD45+ leukocytes	0.5 (0–30)	13 (1–66)	<0.001*
CD3+ T cells	0 (0–4)	0 (0–26)	0.015*

^aMann-Whitney U test used, median and range are given for continuous variables. Asterisk, P value <0.05 considered significant.

^b χ^2 test used, proportions are given.

^cData for histological variables in part from Frösen et al. (7).

^dNo. of cells per standardized grid area, 0.0625 mm².

Of the 58 SCAAs studied for C activation, wall histology and inflammatory cell infiltrations of 43 SCAAs were previously stained and evaluated (6) using histological staining and immunohistochemistry for endothelium (CD31), cell proliferation (Ki67), leukocytes (CD45), T cells (CD3), and macrophages (CD68, CD163 and CD11b).

Electron Microscopy

The SCAA sample was immersion fixed in 4% para-formaldehyde (Electron Microscopy Sciences, Fort Washington, PA). For the morphological analysis, specimens were further fixed in 2.5% glutaraldehyde (Fluka, Buchs, Switzerland), post-fixed with 1% osmium tetroxide, dehydrated, and embedded in Epon (TAAB Laboratories Equipment Ltd., Berkshire, England). The diluent was 100 mmol/L sodium phosphate buffer, pH 7.4.

For immuno-electron microscopy (IEM), para-formaldehyde-only-fixed specimens were low-temperature embedded into Lowicryl (HM20; TAAB Laboratories Equipment Ltd.). Dehydration, resin infiltration, and ultraviolet-polymerization were performed at -30°C . The ultrathin sections (60 nm) were blocked with 1% bovine serum albumin, 0.5% cold fish-skin gelatin (Sigma) and 1% heat inactivated fetal calf serum (BioWhittaker, Walkersville, MD). The primary antibody (A239) was detected with goat anti-mouse IgG and IgM conjugated with 10 nm colloidal gold (British Biocell, Cardiff, England), and both antibodies were diluted with 2% bovine serum albumin, 0.1% Tween 20 (Electron Microscopy Sciences), and 0.1% fish-skin gelatin. All specimens were poststained with uranyl acetate (Ultrastain I; Leica, Wetzlar, Germany), lead citrate (Ultrastain II; Leica), and viewed with a Jeol EX 1200 II transmission electron microscope (Jeol Ltd., Tokyo, Japan) operating at 60 kV.

Histological Evaluation

The positively stained areas of immunoperoxidase-stained samples were measured three times with KS300

3.0 image processing software (Carl Zeiss GmbH, Oberkochen, Germany).

The percentage of MAC-positive area from the total surface area of one section of each histological sample was measured under $\times 1.25$ magnification. The mean values were used to correlate the area of C activation with the patient's clinical parameters, SCAA rupture status, histological markers, and inflammatory cell infiltrations.

The percentages of MAC-positive areas from the specific wall type-representing areas were measured under $\times 5$ magnification. The areas were classified on the basis of the previously published grading system of four different histological, rupture-related SCAA wall types: 1) an endothelialized wall with linearly organized smooth muscle cells; 2) a thickened wall with disorganized smooth muscle cells; 3) a hypocellular wall with

TABLE 2. The correlation of membrane attack complex-positive area with histological and clinical parameters^a

Variables	Complement activation ^b	
	r_s	<i>P</i> value
Patients		
Age ^c	0.211	0.180
Sex ^d		0.970
Patients with multiple SCAAs ^d		0.165
Patients with previous aneurysmal SAHs ^d		0.264
Fisher's grade of unruptured aneurysm ^e		0.788
Fisher's grade of ruptured aneurysm ^e		0.571
Aneurysms resected for study ^c		
Neck diameter (mm)	0.066	0.680
Fundus width (mm)	0.083	0.599
Fundus length (mm)	0.012	0.942
No. of ruptures	0.341	0.027*
Time from rupture to sample resection	0.137	0.440
Histology of SCAA wall ^{d,f}		
Endothelium totally absent		<0.001*
Elastic lamina remnants		0.986
Intraluminal fresh thrombosis		0.108
Intraluminal organizing thrombosis		0.056
Atherosclerotic calcifications		0.207
Pads of intimal hyperplasia		0.458
Cell proliferation		0.223
Inflammatory cell infiltrations ^{c,g}		
CD11b+ macrophages	0.293	0.066
CD68+ macrophages	0.129	0.429
CD163+ macrophages	0.360	0.023*
CD45+ leukocytes	0.280	0.084
CD3+ T cells	0.344	0.030*
Proportion of TUNEL+ cells ^c	0.167	0.309

^a r_s , Spearman-Rank correlation test. Asterisk, **P* value <0.05 considered significant.

^b Relative area positive for MAC.

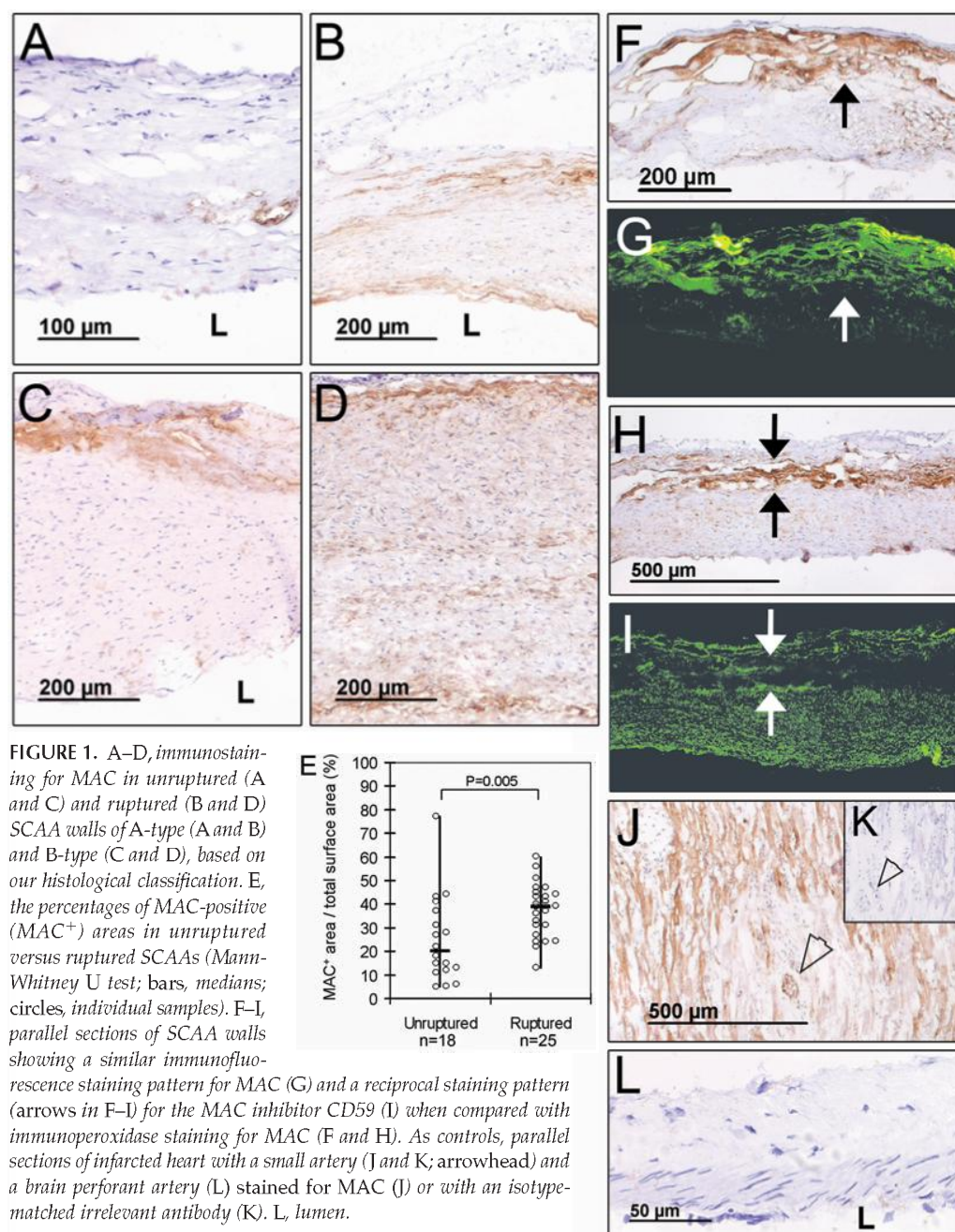
^c Spearman's coefficient of correlation shown where appropriate.

^d Mann-Whitney *U* test used.

^e Kruskal-Wallis test used.

^f Data for histological variables in part from Frösen J et al. (7).

^g No. of cells per standardized grid area, 0.0625 mm².



either myointimal hyperplasia or organized thrombus; and 4) an extremely thin thrombosis-lined hypocellular wall (6).

The histological data for inflammatory cell infiltrations of the 43 previously studied SCAAs has been published by Frösen et al. (6). The inflammatory cell infiltrations were quantified as the number of positive cells, and the proliferation was quantified as the proportion of Ki67-positive nuclei per standardized grid area (0.0625 mm²). The presence or absence of histological parameters listed in *Tables 1* and *2* was categorically scored

(yes or no). The evaluation was performed by two observers (RT, JF). Of these samples, the data were not available in six ruptured SCAAs because of poor orientation of histological sections or nonrepresentative stainings.

The number of TUNEL-positive cells per field of view (0.0956 mm²) and the TUNEL and MAC colocalization were evaluated by two observers (RT, JF) with an Axiovision fluorescence microscope and image processing software (Carl Zeiss).

Statistics

We report proportions, medians, and ranges for clinical and histological variables on patients with unruptured and ruptured samples. Differences were tested with a χ^2 , Mann-Whitney U, Spearman-Rank correlation, or Kruskal-Wallis test, as indicated in *Tables 1* and *2*. Confidence intervals (CI) of 95% for the median were calculated using distribution free quantile estimator and Jack-knife standard error (9).

The association between SCAA wall type and the presence of MAC was analyzed using the square root transformation of the percentage of MAC-positive areas. A linear mixed model was used to account for variation between histological cuts with a varying number of MAC-measurements in each cut. The linear mixed model (13) for the percentage of MAC-positive area in the histological section *i* at measurement *j* is:

between histological cuts with a varying number of MAC-measurements in each cut. The linear mixed model (13) for the percentage of MAC-positive area in the histological section *i* at measurement *j* is:

$$y_{ij} = \alpha_1 + \sum_{p=2}^4 \alpha_p * x_{pij} + \beta_{Li} + \epsilon_{ij}$$

Here α_1 is the mean response in wall Type A, adjusted for other effects, coefficients for fixed effects, and random effects. In all models, α_2 , α_3 , and α_4 are the regression coefficients for

comparing three different SCAA wall types with variables: $x_{2ij} = 1$ if "Type B" wall, 0 otherwise; $x_{3ij} = 1$ if "Type C" wall, 0 otherwise; and $x_{4ij} = 1$ if "Type D" wall, 0 otherwise to "Type A" wall. To allow cut-specific variation in the percentage of MAC-positive area, we introduce a random intercept:

$$\beta_{1i} \stackrel{iid}{\sim} N(0, \sigma_i^2)$$

Basic univariate statistics and tests were obtained with NCSS 2000 (NCSS Statistical Software, Kaysville, UT), the linear mixed model was fitted, and CIs for the medians were calculated using R 2.1.0-A Language and Environment (2).

RESULTS

Patients

The patients with unruptured ($n = 26$) and ruptured ($n = 32$) SCAAs did not differ in age, sex, number of SCAAs, previous SAHs, or in the dimensions of the SCAA. Of the unruptured samples, 21 out of 25 were Fisher Grade 0, and four out of 25 were Grade 3 or 4 because of SAH from SCAAs other than the studied SCAAs. Of the ruptured samples, 21 out of 30 were Fisher Grade 4, six out of 30 were Grade 3, and three out of 30 were less than Grade 3. Of the histological parameters, deendothelialization, organized thrombosis, proliferation ratio, and inflammatory cell infiltrations correlated with the rupture status (Table 1).

C Activation in Unruptured and Ruptured SCAAs Compared with Wall Histology

All SCAA samples stained positive for MAC in certain regions regardless of the rupture status (Fig. 1). Immunoperoxidase and immunofluorescence staining showed similar patterns, and the MAC inhibitor, CD59, stained reciprocally to

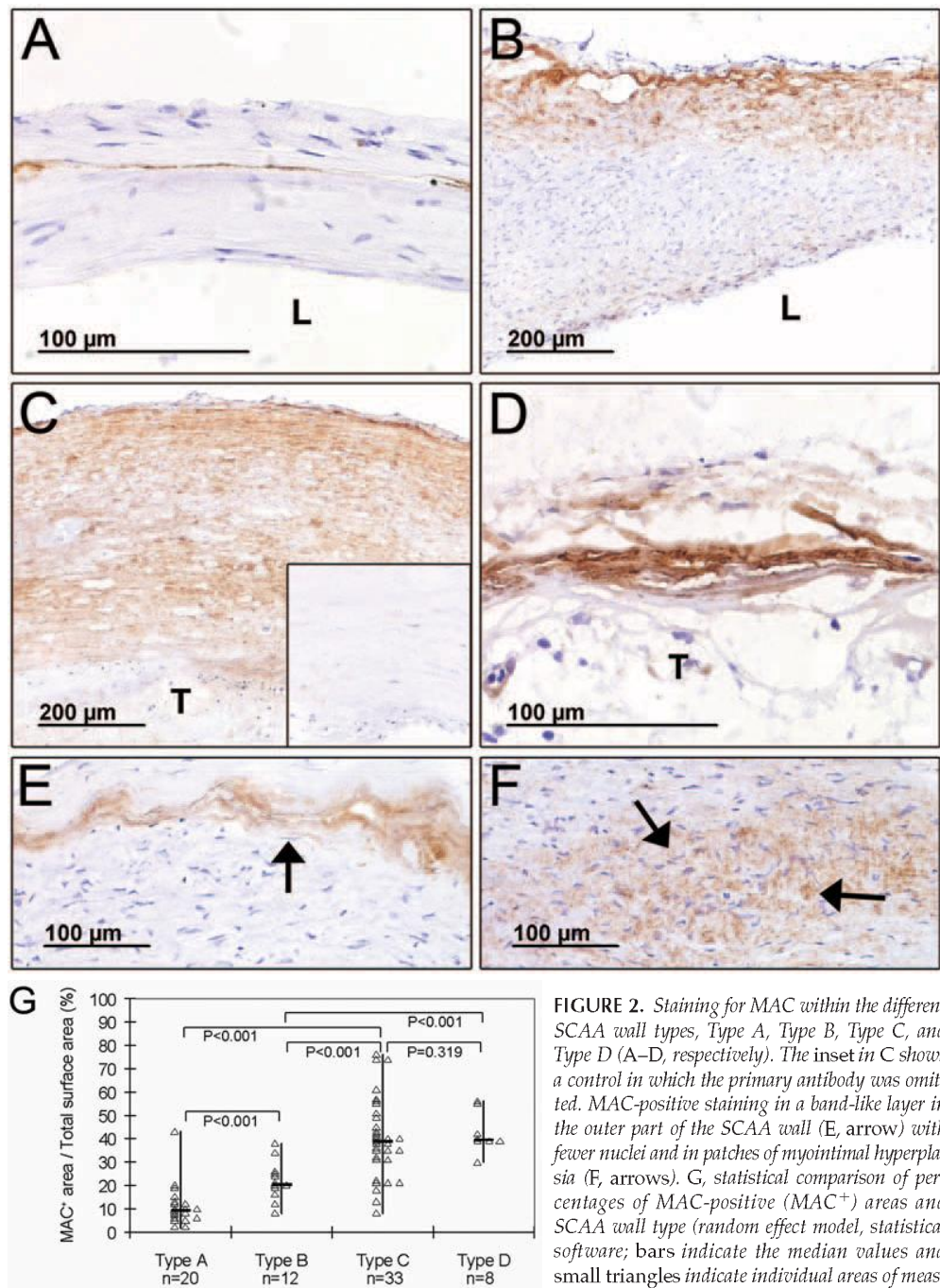


FIGURE 2. Staining for MAC within the different SCAA wall types, Type A, Type B, Type C, and Type D (A–D, respectively). The inset in C shows a control in which the primary antibody was omitted. MAC-positive staining in a band-like layer in the outer part of the SCAA wall (E, arrow) with fewer nuclei and in patches of myointimal hyperplasia (F, arrows). G, statistical comparison of percentages of MAC-positive (MAC⁺) areas and SCAA wall type (random effect model, statistical software; bars indicate the median values and small triangles indicate individual areas of measurement). L, lumen; T, thrombus.

MAC (Fig. 1). MAC depositions localized consistently in a band-like region in the outer SCAA wall (Fig. 2). Fewer nuclei were present on this layer compared with the adjacent MAC-negative area in 28 (80%) out of 35 samples (Fig. 2). Some additional MACs were found occasionally in patches of myointimal hyperplasia. Within these areas, the nuclei were scattered normally in

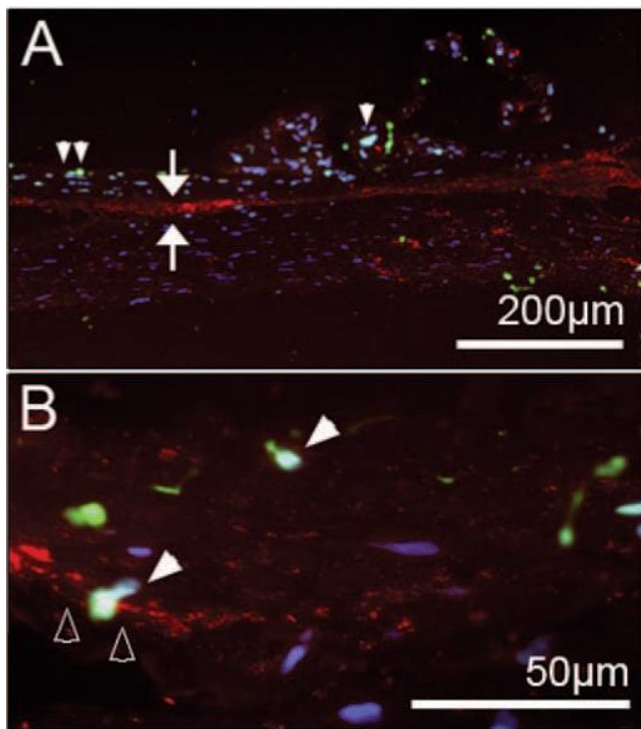


FIGURE 3. MAC (red) and TUNEL (green) double-fluorescence staining, with blue nuclei. **A**, at low power ($\times 5$) magnification, TUNEL-positive nuclei (arrowheads) were found in the luminal or adventitial areas, whereas MAC-positive staining was detected in the inner wall (arrows). **B**, MAC-positive staining (open arrowheads) near the TUNEL-positive nuclei (white arrowheads) at a higher ($\times 40$) magnification on the same unruptured SCAA wall. No double-positive cells (yellow) were detectable.

21 (91%) out of 23 samples (Fig. 2). The median MAC-positive area was 39% (range, 13–60%) in ruptured and 20% (range, 5–77%) in unruptured SCAAs ($P = 0.005$; Fig. 1). Of these samples, seven out of 28 were collected longer than 48 hours after aneurysm rupture, but the MAC-positive areas (median, 39%; range, 13–56%; 95% CI, 35.5–42.7) were comparable to samples collected within 48 hours of the rupture (median, 39%; range, 5–60%; 95% CI, 10.5–31.4). No correlation between MAC-positive areas and Fisher grade was observed in unruptured or ruptured SCAAs (Table 2).

The median percentages of MAC-positive areas within different SCAA wall types were 9% in A-type ($n = 12$), 20.5% in B-type ($n = 12$), 39% in C-type ($n = 33$), and 40% in D type ($n = 9$) walls (Fig. 2). The percentages of areas positive for MAC differed significantly ($P < 0.001$) between all wall types, except between the C- and D-types (Fig. 2).

Of inflammatory cells, only the number of CD163⁺ macrophages ($P = 0.037$) and CD3⁺ T cells ($P = 0.047$) correlated with MAC deposition. However, the significance levels were 0.339 and 0.324, respectively. Of other parameters, the MAC-positive area correlated significantly with de-endothelialization ($P < 0.001$), but not with cell proliferation or thrombosis (Table 2).

C Activation and Cell Death

In double-fluorescence stainings, the TUNEL-positive cells were located mainly on the luminal and adventitial sides, whereas MAC was detected inside the SCAA wall (Fig. 3). Twelve (54%) out of 22 unruptured and 24 (77%) out of 31 ruptured SCAA walls harbored TUNEL-positive cells. The proportion varied between 0 and 65% (median, 11%) in unruptured and between 0 and 89% (median, 26%) in ruptured samples ($P = 0.052$). TUNEL-positive cells colocalized with MAC-positive areas in none out of seven unruptured and four out of seven ruptured samples under $\times 5$ magnification. At the cellular level ($\times 40$ magnification), some C depositions were found adjacent to the TUNEL-positive nuclei in five out of 14 samples, but no double-positive cells could be detected.

In EM, a band-like region of degenerated tissue in the outer part of SCAA wall contained cellular debris, membranous structures, and scattered disorganized collagen bundles rather than organized tissue (Fig. 4). The few cells found were small and crumpled, contrary to the linearly organized cells with intact nuclei and cell organelles in the inner part of the SCAA wall, and, furthermore, showed universal signs of cell death: swollen mitochondria, secondary lysosomes, and tattered membranes. Although some nuclei had condensed chromatin, no other classic features of apoptosis, such as nuclear blebbing, autophagic structures, or pyknotic cells with condensed cytoplasm, could be observed. In IEM, MAC depositions were detected on band-like degenerated area of the outer wall. At the ultrastructural level, the antibody to MAC binding structures resembled remnants of cell membranes and cellular debris observed in corresponding areas in Epon-embedded sections. No MAC was deposited in the organized connective tissue components, such as collagen.

DISCUSSION

In this study, we showed an association of MAC deposition with SCAA wall degeneration and wall rupture. We also found that increased MAC deposition associated with de-endothelialization, as well as with increased T cell (CD3⁺) and CD163⁺ macrophage infiltration in the SCAA wall. However, MAC deposition did not associate with the infiltration of other types of macrophages (CD68⁺ or CD11b⁺) or leukocytes (CD45⁺) and did not associate with proliferation. We found MAC in all of the studied samples regardless of the SCAA rupture status. This suggests that C activation is not solely a response to the rupture but, rather, seems to be involved within the pathobiology of SCAA wall before rupture.

The degenerated and decellularized wall Types C and D, which are found more often in the ruptured SCAAs, showed larger MAC-positive areas than the proliferative B-type and the quiescent, endothelialized A-type SCAA walls, which are more often unruptured (6). Thus, MAC deposition seems to associate with a weaker, more rupture-prone SCAA wall. We found fewer nuclei in the MAC-positive areas in the outer SCAA wall, which indicates a major cell loss in these regions.

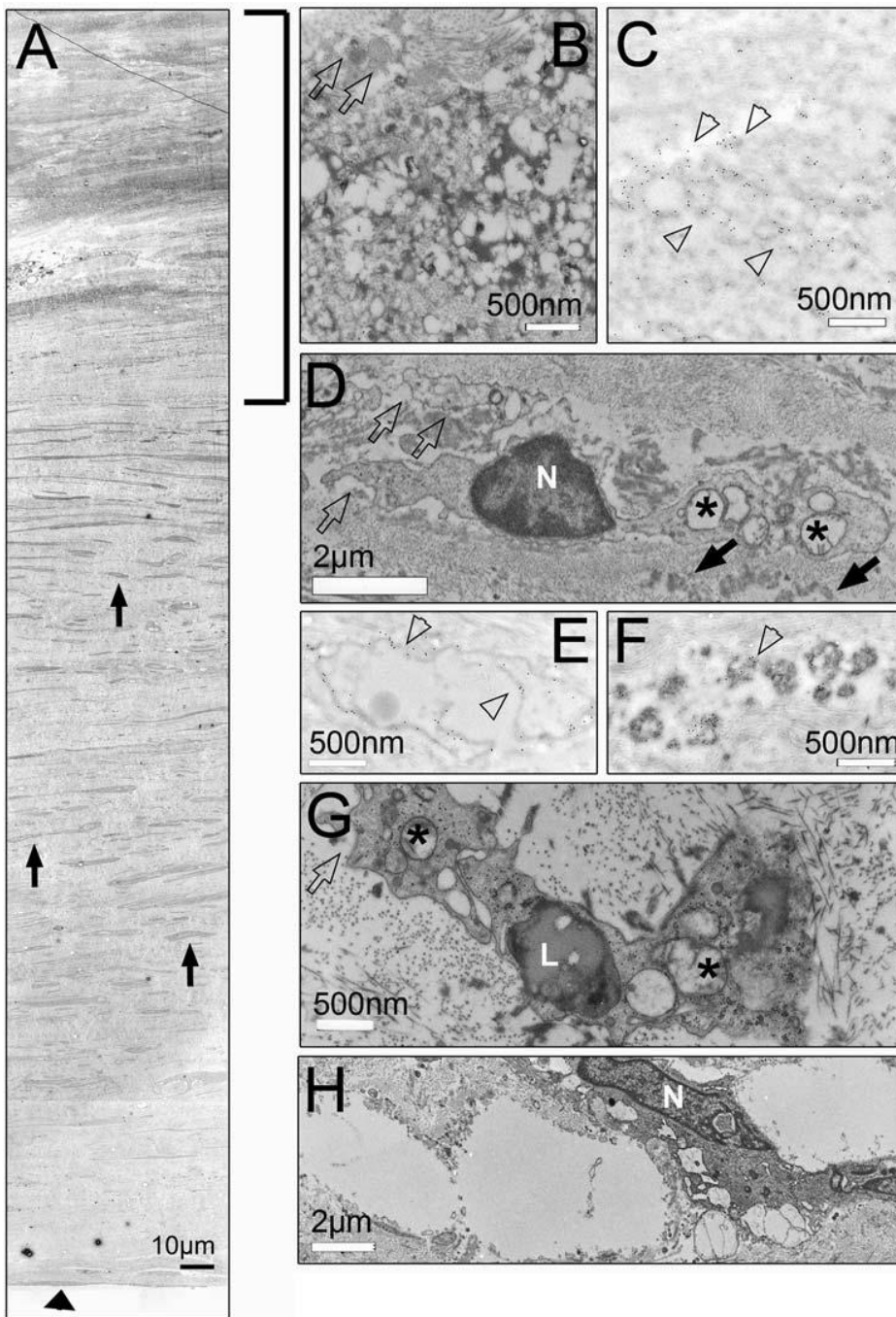


FIGURE 4. EM (epon-embedded sections A, B, D, G, H) and IEM (C, E, F) analysis of cell death and MAC deposition in the SCAA wall of an unruptured SCAA. A, reconstruction image at $\times 800$ magnification illustrates a cross section from endothelium (black arrowhead) to a degenerated area on the outer SCAA wall (bracket). The inner wall contains linearly organized cells (arrows) surrounded by organized connective tissue. B and C, cellular debris and tattered membranes in the degenerated band-like area (open arrow) and the corresponding areas with MAC deposits (open arrowheads). D and G, dying cells with swollen mitochondria (asterisk), secondary lysosomes (L), tattered membranes (open arrow) and nearby granular debris (arrow). The corresponding IEM of cell membrane-resembling (E) and granular debris-resembling (F) structures. (H) Areas of torn tissue within the degenerated band-like area.

However, the causality between C activation and cell death warrants further studies.

Activated C can kill cells by direct lysis or apoptosis, but C can also get activated as an endogenous clearance mechanism that removes senescent or damaged cells (3, 5). An increase in apoptotic cell death in ruptured SCAAs has recently been described (6, 7, 16, 19). However, we found only a weak correlation between TUNEL-positive cells and MAC deposition. This suggests that apoptosis is not the major mechanism behind C activation and cell loss, despite the occasional findings of MAC near TUNEL-positive cells. The rarity of colocalization of MAC and TUNEL-positive cells suggests that C-mediated cell death and apoptosis could have their own distinct roles in the cell loss. An intriguing alternative hypothesis is that C5b-9 complex may have protective effects against apoptosis in SCAA wall. Previous studies have shown that sublytic concentrations of MAC can protect from apoptotic cell death via regulation of intracellular pathways and growth factor production (2, 18, 23).

In addition, the clotting cascade and fibrinolysis have been shown to activate the C system, e.g., via plasmin-induced cleavage of C3 (10, 22). In our series, the ruptured samples showed more organized thrombosis compared with unruptured SCAAs. The comparison of MAC positive area and organized thrombosis showed a trend toward correlation, although it remained statistically nonsignificant. However, whether thrombosis induces MAC formation in the SCAA wall or whether their presence is coincidental demands further studies.

C activation is chemotactic for inflammatory cells (20), and C activation triggered by foreign surfaces, e.g., those of microbes, is usually associated with an apparent inflammatory cell infiltration. The lack of correlation between MAC deposition and infiltration of inflammatory

cells other than T cells and CD163⁺ macrophages suggests an absence of an infectious focus. Nevertheless, the possible relation between C activation in SCAA wall and other types of antigen-specific inflammatory responses cannot be excluded. Furthermore, the alternatively activated CD163⁺ macrophages are more likely linked to the dampening of the inflammatory response and resolution of inflammation than to an acute inflammatory reaction associated with direct killing of pathogens (14). Whether or not they are related to C-directed phagocytosis remains to be studied.

CONCLUSIONS

We found that C activation, featured as MAC deposition, associates with SCAA wall degeneration and rupture and localizes mainly to the outer media-representing region of the SCAA wall. The possible role of C activation in the SCAA wall degeneration warrants further studies to elucidate the role of C activation in the pathobiology of the SCAA wall.

REFERENCES

- Anonymous: The R-Project for Statistical Computing. <http://www.R-project.org>. Accessed 1/20/06.
- Bohana-Kashtan O, Ziporen L, Donin N, Kraus S, Fishelson Z: Cell signals transduced by complement. *Mol Immunol* 41:583–597, 2004.
- Chyatte D, Bruno G, Desai S, Todor DR: Inflammation and intracranial aneurysms. *Neurosurgery* 45:1137–1147, 1999.
- Cole DS, Morgan BP: Beyond lysis: how complement influences cell fate. *Clin Sci (Lond)* 104:455–466, 2003.
- Fisher CM, Kistler JP, Davis JM: Relation of cerebral vasospasm to subarachnoid hemorrhage visualized by computerized tomographic scanning. *Neurosurgery* 6:1–9, 1980.
- Frösen J, Piippo A, Paetau A, Kangasniemi M, Niemelä M, Hernesniemi JA, Jääskeläinen J: Remodeling of saccular cerebral artery aneurysm wall is associated with rupture: Histological analysis of 24 unruptured and 42 ruptured cases. *Stroke* 35:2287–2293, 2004.
- Hara A, Yoshimi N, Mori H: Evidence for apoptosis in human intracranial aneurysms. *Neurol Res* 20:127–130, 1998.
- Harrell FE, Davis CE: A new distribution-free quantile estimator. *Biometrika* 69:635–640, 1982.
- International Study of Unruptured Intracranial Aneurysms Investigators: Unruptured intracranial aneurysms-risk of rupture and risks of surgical intervention. *N Engl J Med* 339:1725–1733, 1998.
- Johnson RJ: Complement activation during extracorporeal therapy: Biochemistry, cell biology and clinical relevance. *Nephrol Dial Transplant* 9:36–45, 1994.
- Kataoka K, Taneda M, Asai T, Kinoshita A, Ito M, Kuroda R: Structural fragility and inflammatory response of ruptured cerebral aneurysms. A comparative study between ruptured and unruptured cerebral aneurysms. *Stroke* 30:1396–1401, 1999.
- Kitagawa M, Yamaguchi S, Takahashi M, Tanizawa T, Hirokawa K, Kamiyama R: Localization of Fas and Fas ligand in bone marrow cells demonstrating myelodysplasia. *Leukemia* 12:486–492, 1998.
- Laird NM, Ware JH: Random-effects models for longitudinal data. *Biometrics* 38:963–974, 1982.
- Moestrup SK, Møller HJ: CD163: A regulated hemoglobin scavenger receptor with a role in the anti-inflammatory response. *Ann Med* 36:347–354, 2004.
- Oksjoki R, Jarva H, Kovanen PT, Laine P, Meri S, Pentikäinen MO: Association between complement factor H and proteoglycans in early human coronary atherosclerotic lesions: Implications for local regulation of complement activation. *Arterioscler Thromb Vasc Biol* 23:630–636, 2003.
- Pentimalli L, Modesti A, Vignati A, Marchese E, Albanese A, Di Rocco F, Coletti A, Di Nardo P, Fantini C, Tirpakova B, Maira G: Role of apoptosis in intracranial aneurysm rupture. *J Neurosurg* 101:1018–1025, 2004.
- Ronkainen A, Miettinen H, Karkola K, Papinaho S, Vanninen R, Puranen M, Hernesniemi JA: Risk of harboring an unruptured intracranial aneurysm. *Stroke* 29:359–362, 1998.
- Rus HG, Niculescu FI, Shin ML: Role of the C5b-9 complement complex in cell cycle and apoptosis. *Immunol Rev* 180:49–55, 2001.
- Sakaki T, Kohmura E, Kishiguchi T, Yuguchi T, Yamashita T, Hayakawa T: Loss and apoptosis of smooth muscle cells in intracranial aneurysms. Studies with in situ DNA end labeling and antibody against single-stranded DNA. *Acta Neurochir (Wien)* 139:469–475, 1997.
- Sim RB, Tsiftoglou SA: Proteases of the complement system. *Biochem Soc Trans* 32:21–27, 2004.
- van Gijn J, Rinkel GJ: Subarachnoid haemorrhage: diagnosis, causes and management. *Brain* 124:249–278, 2001.
- Ward PA: A plasmin-split fragment of C'3 as a new chemotactic factor. *J Exp Med* 126:189–206, 1967.
- Zwaka TP, Torzewski J, Hoeflich A, Dejosez M, Kaiser S, Hombach V, Jähle PM: The terminal complement complex inhibits apoptosis in vascular smooth muscle cells by activating an autocrine IGF-1 loop. *FASEB J* 17:1346–1348, 2003.

Acknowledgments

We thank Ilse Lappalainen, Tanja Eriksson, and Mervi Lindman for their technical assistance and Maarit Alalahti for her help with sample collection. This work was supported by Helsinki University Central Hospital research funds (EVO TYH4204, TYH5204, and TYH5233); the Sigrid Jusélius Foundation, Helsinki, Finland; The Academy of Finland; the Maine Taponen Foundation, Finland; and the Alcohol Research Foundation, Helsinki, Finland. We certify that we have no affiliation with or financial involvement in any organization or entity with a direct financial interest in the subject matter or materials discussed in this study.

COMMENTS

This is another thought provoking study from the group in Helsinki, indicating that complement activation is a reaction within the aneurysm wall, which leads to saccular aneurysm rupture. In a series of 26 unruptured and 32 ruptured aneurysms, they examined histological indicators of complement activation, macrophage and leukocyte infiltration, endothelial status, and luminal thrombosis. In an area of cell loss in the aneurysm wall, they demonstrated significantly increased complement activation in the ruptured aneurysm cohort.

The two groups of aneurysm patients seemed to be well matched, and there were no significant clinical differences in terms of their age, sex, number of aneurysms, previous subarachnoid hemorrhage (SAH), or the dimensions of the aneurysms. It is unclear whether or not the unruptured group had a higher or lower incidence of hypertension. The authors considered the possibility that the increased complement activation could have been caused solely by the rupture itself, but they concluded that this was not the case, as certain types of macrophage and leukocyte infiltrations, cellular proliferation, and luminal thrombosis were not found in the ruptured aneurysm group.

These findings support the hypothesis previously proposed by the authors and others that aneurysms that are prone to rupture are fundamentally different from those that are destined to remain unruptured. It is likely that a set of mechanisms within the aneurysm and vessel wall and related to systemic hemodynamics may separate aneurysms that are likely to rupture from those that will not. Clearly, aneurysm size is one determinant factor in the development of rupture, but this and other work indicates that there is more happening here than a simple size-related propensity to rupture.

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This study reports observations on resected aneurysm sacs from the vastly experienced group in Helsinki, Finland. They report an association between the membrane attack complex (MAC) and aneurysm wall degeneration. The observations are interesting, and the argument that complement activation and inflammation contribute to the degeneration is, at the least, logical. The work is exquisitely well illustrated and presented. One problem is that the range of MAC-positive areas for ruptured and unruptured aneurysms is huge and they overlap, even though a statistical test was discovered that could produce a "significant" difference. Whether or not this is clinically important is unknown.

The second and key fundamental problem remains that one wants to know the findings in the ruptured aneurysm before it ruptures and then compare that with the aneurysm that never ruptured. Studies like this show only association, and one can never know anything about cause and effect. Also, some of the unruptured aneurysms were in patients who had SAH. In the absence of an animal model of aneurysm formation and rupture, however, this may be the best that can be done. The only other thing that might help would be examining the time course of changes in ruptured aneurysms. This series had aneurysm samples from 4 hours to 47 days after SAH. Interesting information might be derived from study of unruptured and ruptured aneurysms from the same patient because the former may have a higher risk of subsequent rupture than unruptured aneurysms in patients who have never had SAH. The findings also need to account for the strongest factor associated with rupture, i.e., increasing aneurysm size. Because size seems to affect hemodynamics predominately, figuring out how this affects complement expression might be of interest.

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A cerebral aneurysm (CA) can be a cause of SAH, one of the most catastrophic forms of stroke. Despite its public and social impor-

tance, the mechanism of progression and rupture of CAs has not yet been fully understood. One of the pathological features of CAs is a degenerative change in aneurysmal walls, including the disappearance of internal elastic laminae and the thinning of smooth muscle cell layers. Inflammation was first suggested by Virchow as a cause of CAs, and it is well known that a variety of inflammatory cells accumulate in arterial walls of human CAs. Tulamo et al. demonstrated MAC deposition in aneurysmal walls by immunohistochemistry. MAC deposition was more abundantly seen in rupture-prone aneurysmal walls and correlated with infiltrations of some inflammatory cells and apoptotic cell death. There is no doubt that inflammatory responses in aneurysmal walls are involved in the process of aneurysm progression, eventually resulting in rupture. Although complement activation seems to be associated with wall degeneration and rupture, it is still unclear how it contributes to degenerative changes occurring in aneurysmal walls. MAC deposition may be a result of inflammatory reaction in aneurysmal walls.

We should take a multidisciplinary and comprehensive strategy for investigating molecular mechanisms of progression of CAs. Analysis of human operative samples provides us some important information on the pathogenesis of CAs. However, it is difficult to examine the early stage of CAs by using human samples, as humans have various genetic backgrounds. Other promising approaches are an animal model of CAs and a genetic analysis of CA susceptible genes. Each approach covers the shortcomings of other approaches, and the combination of these three approaches would enable us to clarify the complicated molecular mechanisms of CA progression, thereby developing a novel treatment modality for prevention of aneurysm rupture.

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FUTURE MEETINGS—CONGRESS OF NEUROLOGICAL SURGEONS

The following are the planned sites and dates for future annual meetings of the Congress of Neurological Surgeons:

2007	San Diego, CA	September 15–20
2008	Orlando, FL	September 20–25
2009	New Orleans, LA	October 24–29
2010	San Francisco, CA	October 16–21

FUTURE MEETINGS—AMERICAN ASSOCIATION OF NEUROLOGICAL SURGEONS

The following are the planned sites and dates for future annual meetings of the American Association of Neurological Surgeons:

2007	Washington, DC	April 14–19
2008	Chicago, IL	March 29–April 3
2009	San Diego, CA	May 2–7
2010	Philadelphia, PA	May 1–6